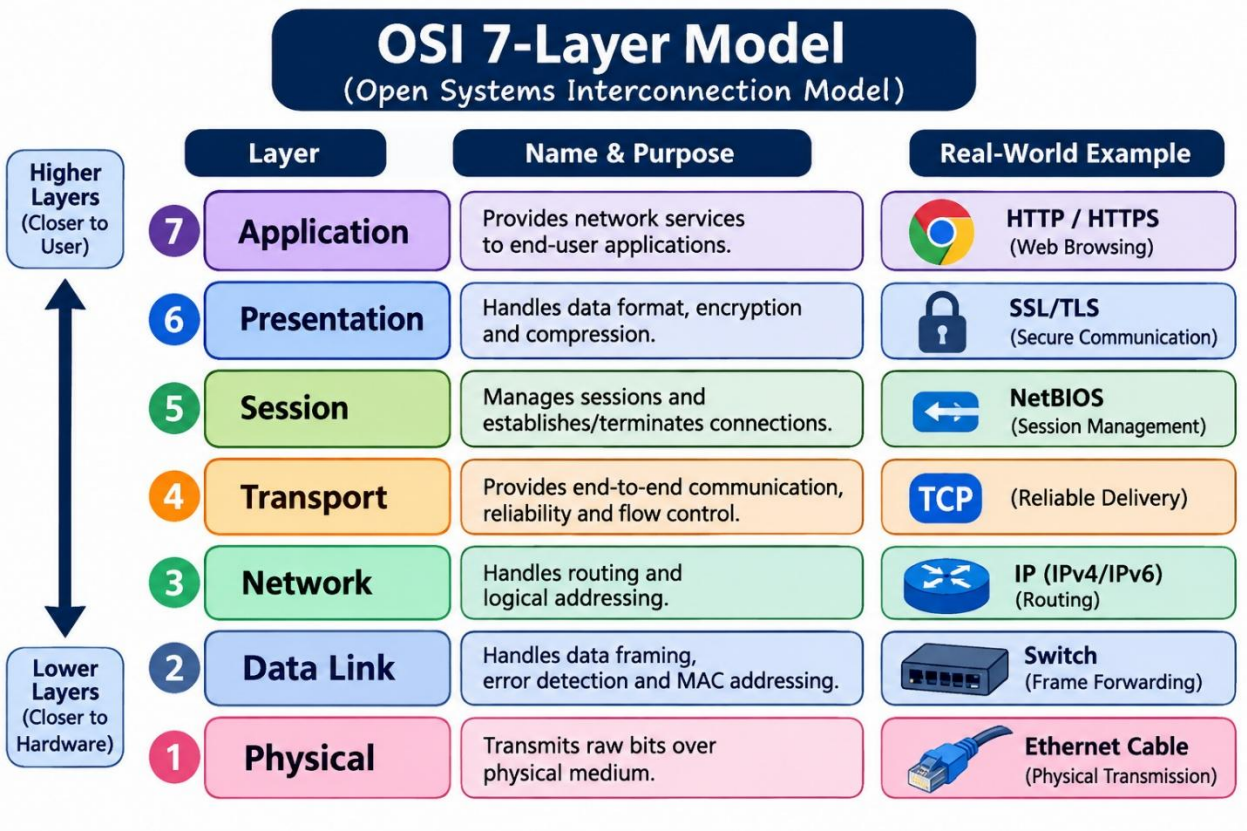
OSI Layer

Tasks 1 : Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data Link, HTTP at Application).



Conclusion: Each OSI layer performs a specific function that helps network devices communicate and transfer data.

Task 2 : Match the following networking scenarios to the correct OSI layer:

a) Encrypting data before sending,

b) Routing packets across networks,

c) Converting data to electrical signals,

d) Opening a web page in your browser,

e) Detecting and correcting errors in frames.

Answer:

a) Encrypting data before sending: **Presentation Layer (Layer 6)**

b) Routing packets across networks: **Network Layer (Layer 3)**

c) Converting data to electrical signals: **Physical Layer (Layer 1)**

d) Opening a web page in your browser: **Application Layer (Layer 7)**

e) Detecting and correcting errors in frames: **Data Link Layer (Layer 2)**

Task 3 : Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.

When I send a message using WhatsApp, all seven layers of the OSI model help explain how data is prepared, transmitted, routed, and received.

1. **Application Layer (Layer 7):** WhatsApp provides the messaging interface and allows users to send and receive messages.
2. **Presentation Layer (Layer 6):** This layer represents data formatting, encoding, and encryption. WhatsApp uses end-to-end encryption to protect personal messages, although its actual implementation does not map perfectly to a separate OSI layer.
3. **Session Layer (Layer 5):** This layer represents the management of communication sessions between applications, including establishing, maintaining, and ending communication.
4. **Transport Layer (Layer 4):** Transport protocols carry data between endpoints. Depending on the service and connection, protocols such as TCP or UDP may be involved.
5. **Network Layer (Layer 3):** IP addressing and routing help packets travel across networks toward their destination.
6. **Data Link Layer (Layer 2):** Wi-Fi or cellular link technology transfers frames between the phone and the local network infrastructure.
7. **Physical Layer (Layer 1):** Radio waves carry data between the phone and the Wi-Fi access point or cellular base station.

Conclusion

All seven OSI layers provide a useful conceptual explanation of communication through WhatsApp. Together, they help us understand how application data is processed and transmitted across a network.

Task 4 : Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.

Problem

I cannot access a website, but I can successfully ping its IP address.

Possible OSI Layers Involved

1. Application Layer (Layer 7): The browser or website application may have a problem, or the website may be unavailable.

2. Transport Layer (Layer 4): The website's TCP port, such as port 443 for HTTPS or port 80 for HTTP, may be blocked or not accepting connections.

3. Presentation Layer (Layer 6): An HTTPS/TLS certificate or encryption negotiation problem may prevent a secure connection.

4. Application Layer – DNS: If the website is accessed by domain name, DNS resolution may be failing even though its IP address responds to ping.

Reasoning

- A successful ping confirms that ICMP packets can reach the IP address and receive replies.
- However, ping does not prove that the website's HTTP or HTTPS service is working.
- The problem may involve DNS, a blocked TCP port, TLS, the browser, or the web server itself.